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| DS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Session | Graph | Question Information | Level 1   Challenge 84 |
| <p><b>Question description</b></p> <p>Byteland has <math>n</math> cities and <math>m</math> roads between them. The goal is to construct new roads so that there is a route between any two cities. Your task is to find out the minimum number of roads required, and also determine which roads should be built.</p> <p><b>Constraints</b></p> $1 \leq n \leq 10^5$ $1 \leq m \leq 2 \cdot 10^5$ $1 \leq a, b \leq n$ <p><b>Input</b></p> <p>The first input line has two integers <math>n</math> and <math>m</math>: the number of cities and roads. The cities are numbered <math>1, 2, \dots, n</math>.</p> <p>After that, there are <math>m</math> lines describing the roads. Each line has two integers <math>a</math> and <math>b</math>: there is a road between those cities.</p> <p>A road always connects two different cities, and there is at most one road between any two cities.</p> <p><b>Output</b></p> <p>First print an integer <math>k</math>: the number of required roads.</p> <p>Then, print <math>k</math> lines that describe the new roads. You can print any valid solution.</p> |         |       |                      |                        |

✓ Logical Test Cases

| Test Case 1                                                                   | Test Case 2                                                                   |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| INPUT (STDIN)<br><br>4 2<br>1 2<br>3 4<br><br>EXPECTED OUTPUT<br><br>1<br>2 4 | INPUT (STDIN)<br><br>4 2<br>2 3<br>2 4<br><br>EXPECTED OUTPUT<br><br>1<br>1 4 |

✓ Mandatory Test Cases

| Test Case 1               |
|---------------------------|
| KEYWORD<br><br>while(m--) |

✓ Complexity Test Cases

| Test Case 1                    | Test Case 2            | Test Case 3    |
|--------------------------------|------------------------|----------------|
| CYCLOMATIC COMPLEXITY<br><br>5 | TOKEN COUNT<br><br>475 | NLOC<br><br>50 |